

Directional Shielding and Noise-Related Loss Characterization of Multilayer Thin-Ferrite Cylindrical Shields for Low-Noise Magnetic Instrumentation

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Abstract—Low-noise magnetic instruments are limited not only by sensor noise, but also by leakage fields through the shield and shield-induced material-loss noise. This internal review draft establishes a measurement-oriented characterization methodology for multilayer thin-ferrite cylindrical shields. The method defines component-wise directional shielding factors using validated no-shield references and uses a separate virtual pickup-coil finite-element model to evaluate the geometry-dependent magnetic-noise-related loss indicator $\int H^2 dV$ (IntH2). A preliminary transverse continuous-shell data chain has been established for candidate geometries with 100 mm inner radius and 200 mm length. The draft also defines the paper-use criteria and validation matrix required before IEEE TIM submission, including mesh and boundary-domain convergence, axial shielding, segmented-shell correction, permeability sensitivity, field-map interpretation, and optional prototype shielding-factor measurement. The current document is an IEEE TIM-oriented internal review draft; it establishes the methodology and data-entry rules and does not make a final instrument-level recommendation.

Index Terms—Ferrite magnetic shield, finite-element method, instrumentation, magnetic noise, multilayer cylindrical shell, shielding factor.

I. INTRODUCTION

ULTRA-SENSITIVE magnetic measurement systems, including atomic magnetometers, spin-exchange-relaxation-free (SERF) comagnetometers, magnetoencephalography instruments, and precision tests of fundamental symmetries, require a controlled low-field and low-noise magnetic environment [1]–[5]. In such systems, the magnetic shield is part of the measurement chain. It determines the environmental-field leakage that reaches the sensor and can also generate magnetic-field noise through material loss. Therefore, the shield should not be evaluated only by a single shielding factor. For low-noise magnetic instrumentation, directional shielding factor, center leakage field, ferrite volume, radial space occupation, and material-loss-related noise indicators should be evaluated together.

High-permeability metallic shields are widely used to attenuate static and low-frequency fields [6]–[8]. However, their high electrical conductivity produces eddy-current magnetic noise that can become relevant in the frequency band of high-sensitivity magnetometers [9]–[11]. Ferrites are attractive

alternatives because their conductivity is much lower than that of metallic alloys. This suppresses eddy-current noise; the remaining low-frequency shield-induced noise is then associated mainly with magnetic loss, represented by the imaginary permeability μ'' , and with the spatial distribution of magnetic field inside the ferrite [9], [12]–[15]. A recent IEEE TIM study on cubic ferrite shields demonstrated the practical importance of ferrites in compact low-noise shielding [16]. Nevertheless, the cubic panel geometry does not directly address open-ended cylindrical shields, which are common around atomic sensors and compact comagnetometers.

A monolithic cylindrical ferrite shield is mechanically difficult to fabricate when the diameter is large and the wall is thin. Ferrites are brittle after sintering, and machining a 200-mm-diameter thin-walled cylinder with sub-millimeter wall thickness is costly and fragile. A manufacturable alternative is to assemble a shield from multilayer thin-ferrite cylindrical shells or segmented ferrite sheets. This introduces a coupled design problem: layer number N_L , ferrite thickness a , radial gap g , circumferential segmentation, ferrite volume V_f , and total radial occupation all affect the center leakage field and the geometry-dependent loss integral used in magnetic-noise calculations.

Existing studies provide important foundations, but do not fully cover this measurement-oriented design problem. Kornack *et al.* reported a low-noise ferrite cylinder with measured magnetic noise [12]. Lee and Romalis derived magnetic-noise formulas for high-permeability shields and conducting objects with simple geometry [9]. Later studies modeled cylindrical or multi-annular ferrite shields and discussed gap effects and material parameters [13], [14], [17]. However, previous work has not simultaneously reported a validated no-shield directional reference, layer-resolved IntH2 analysis, and a multilayer thin-ferrite cylindrical geometry under a unified measurement-oriented characterization process. Table I summarizes the relationship between representative prior work and the present manuscript.

The main contributions of the current internal-review draft are: 1) a component-wise directional shielding-factor characterization method based on validated no-shield references; 2) a virtual pickup-coil IntH2 characterization method for the geometry-dependent part of shield-induced material-loss noise; 3) a layer-resolved loss-integral decomposition for multilayer cylindrical ferrite shields; 4) a fixed-volume comparison

TABLE I
RELATIONSHIP TO REPRESENTATIVE FERRITE SHIELD STUDIES

Work	Geometry	Main quantity	Scope relative to this work
Kornack <i>et al.</i> [12]	Single ferrite cylinder	SF and measured noise	No multilayer thin-shell comparison
Lee and Romalis [9]	Simple analytic shapes	Magnetic-noise formula	No finite-length multilayer FEM geometry
Ma <i>et al.</i> [13]	Cylindrical ferrite shield	SF and IntH2	Layer-resolved IntH2 not reported
Lu <i>et al.</i> [14], [17]	Multi-annular/gapped ferrite shields	Noise-related FEM analysis	Directional no-shield reference not emphasized
Sun <i>et al.</i> [16]	Cubic ferrite shield	SF and measured noise	Cubic geometry, not cylindrical multilayer shell
This work	Multilayer thin-ferrite cylinder	Directional SF, V_f , layer-wise IntH2	Measurement-oriented characterization method with validation studies

Fig. 1. Measurement-oriented evaluation framework

External uniform-field model and virtual pickup-coil model. Replace with vector artwork before submission.

Fig. 1. Measurement-oriented characterization principle. The external uniform-field problem provides B_0 , B_{center} , SF_q , and LeakageRatio_q ; the virtual pickup-coil problem provides IntH2 and layer-resolved IntH2.

logic for single-layer and multilayer thin-ferrite candidates; and 5) a validation matrix defining the remaining simulations and experiments required before IEEE TIM submission.

II. MEASUREMENT-ORIENTED CHARACTERIZATION METHODOLOGY

A. Coordinate System and Measurement Configuration

The shield is modeled as a device under test (DUT) placed at the center of a calibrated magnetic-field region. The cylinder axis is defined as the Cartesian z -axis, while x and y span the transverse plane. Because the shield is a finite open-ended cylinder, transverse and axial shielding are not equivalent. The quantities SF_x and SF_y describe transverse shielding through the sidewall, whereas SF_z describes axial shielding affected by leakage through the open ends [6], [27]–[29].

Two finite-element models are used, as shown schematically in Fig. 1. These models answer different measurement questions and should not be physically conflated. The external uniform-field model evaluates environmental-field attenuation and provides B_0 , B_{center} , SF_q , and LeakageRatio_q . The virtual pickup-coil model evaluates how material loss couples to the measured magnetic field through the fluctuation-dissipation theorem and provides $\text{IntH2}_{\text{total}}$ and IntH2_{L_i} .

B. Directional Shielding Factor and Leakage Ratio

For an external field applied along $q \in \{x, y, z\}$, the no-shield reference field at the observation point is

$$\mathbf{B}_0 = (B_{0,x}, B_{0,y}, B_{0,z}), \quad (1)$$

and the shielded center leakage field is

$$\mathbf{B}_{\text{center}} = (B_{\text{center},x}, B_{\text{center},y}, B_{\text{center},z}). \quad (2)$$

The directional shielding factor is defined by

$$SF_q = \frac{|B_{0,q}|}{|B_{\text{center},q}|}, \quad (3)$$

and the corresponding leakage ratio is

$$\text{LeakageRatio}_q = \frac{|B_{\text{center},q}|}{|B_{0,q}|} = \frac{1}{SF_q}. \quad (4)$$

The formal shielding factor is computed from the intended directional component. The vector magnitude B_{Mag} is retained only as an auxiliary consistency check and is not used as the primary SF definition.

C. No-Shield Directional Reference

The no-shield model is the calibration reference for the external-field problem. The ferrite bodies are deleted or assigned vacuum while the air region, boundary dimensions, observation point, solver setting, and excitation amplitude are kept identical to the shielded model. In vacuum, for $H_0 = 1 \text{ A/m}$,

$$B_0 = \mu_0 H_0 = (4\pi \times 10^{-7})(1) = 1.256637 \times 10^{-6} \text{ T}. \quad (5)$$

Thus, an ideal x -directed reference should give $B_{0,x} \approx 1.256637 \times 10^{-6} \text{ T}$, with $B_{0,y}$ and $B_{0,z}$ much smaller. The residual transverse components are treated as numerical residuals caused by finite-element discretization, finite-domain boundary approximation, and interpolation at the observation point.

The reference field is accepted only if the intended component dominates the other components:

$$\Gamma_q = \frac{|B_{0,q}|}{\max_{p \neq q} |B_{0,p}|}. \quad (6)$$

In the post-processing rule set, a failed directional reference prevents the corresponding shielding factors from being used in figures or conclusions.

D. Analytical Scaling Motivation

The finite-element model is the primary evaluation tool because the shield is finite in length, open ended, and may include layer gaps or segmented slots. Nevertheless, the choice of design variables is guided by the classical transverse

cylindrical-shell solution. In a source-free region with linear isotropic permeability, $\nabla \times \mathbf{H} = 0$ and $\nabla \cdot \mathbf{B} = 0$ allow the magnetic scalar potential to satisfy Laplace's equation. For an infinitely long coaxial cylindrical shell in a transverse uniform field, the potential in each annular region can be written as

$$\phi_k(r, \theta) = \left(C_k r + \frac{D_k}{r} \right) \cos \theta, \quad (7)$$

where the constants are obtained from the continuity of normal B and tangential H at each interface [6]. For a single high-permeability thin shell, this solution gives the familiar leading transverse scaling

$$S_T \simeq 1 + \frac{\mu_r t}{2R}, \quad (8)$$

showing that shell thickness, mean radius, and permeability set the first-order shielding contribution.

For a multilayer coaxial shell, the analytical expansion contains direct layer terms and interlayer radius-coupling terms. A typical coupling factor between two layers with mean radii $R_i < R_j$ is

$$\Lambda_{ij} = 1 - \left(\frac{R_i}{R_j} \right)^2. \quad (9)$$

This motivates comparing layer number, radial gap, and total radial occupation rather than comparing total thickness alone. The present manuscript uses this analytical structure only as a scaling motivation; the reported SF_q values are obtained from the validated directional FEM reference in (3), not from the ideal infinite-cylinder expression.

Under the thin-shell approximation, a set of equal-height coaxial layers with total ferrite thickness $T_f = N_L a$ has

$$V_f \approx 2\pi L R_c T_f, \quad (10)$$

where R_c is the radial-window center radius. Therefore, fixed T_f is a useful first-order proxy for fixed ferrite volume. In the actual post-processing, however, V_f is always computed from the exact layer radii in (22) or from CAD volume export.

E. Virtual Pickup-Coil Model and IntH2

The shield-induced magnetic noise can be related to material loss through the fluctuation-dissipation theorem [9], [18]. A virtual pickup coil placed at the observation point produces a magnetic field H_{vc} in the ferrite. For coil area A , turn number N_c , peak current I_0 , and magnetic moment $m = AN_c I_0$, the hysteresis-loss contribution to the magnetic-field noise amplitude spectral density can be written as

$$\delta B_{\text{hyst}}(f) = \frac{1}{m} \left(\frac{2k_B T \mu''}{\pi f} \int_V H_{vc}^2 dV \right)^{1/2}, \quad (11)$$

where $\mu'' = \mu_0 \mu_r''$ is the absolute imaginary permeability if μ_r'' is specified as a relative quantity. The current I_0 is a peak current; if an RMS current is used in another implementation, the normalization must be converted consistently.

This manuscript does not claim measured magnetic-noise spectra. The exported FEM quantity is the geometry-dependent magnetic-noise-related loss indicator

$$\text{IntH2} = \int_V H_{vc}^2 dV, \quad (12)$$

with SI unit $A^2 m$. IntH2 becomes an absolute noise prediction only after measured $\mu_r''(f)$, temperature, frequency, and virtual-coil normalization are specified. Relative comparison by IntH2 is meaningful only for identical material properties, identical coil normalization, and identical observation direction.

For two candidate geometries A and B evaluated under identical material loss and virtual-coil normalization, (11) gives the relative loss-induced field-noise scaling

$$\frac{\delta B_A(f)}{\delta B_B(f)} = \left(\frac{\text{IntH2}_A}{\text{IntH2}_B} \right)^{1/2}. \quad (13)$$

Thus, IntH2 ratios should not be interpreted linearly as magnetic-noise ratios; the square-root relation applies only under the identical-normalization assumptions stated above.

For a multilayer shield, each ferrite layer is retained as an independent body. The layer-resolved terms are

$$\text{IntH2}_{L_i} = \int_{V_{L_i}} H_{vc}^2 dV, \quad (14)$$

and

$$\text{IntH2}_{\text{total}} = \sum_{i=1}^{N_L} \text{IntH2}_{L_i}. \quad (15)$$

The export is accepted only if

$$\frac{|\text{IntH2}_{\text{total}} - \sum_i \text{IntH2}_{L_i}|}{|\text{IntH2}_{\text{total}}|} < 10^{-3}. \quad (16)$$

F. Multilayer Cylindrical Geometry

The DUT is an open-ended multilayer cylindrical shell with inner radius R_{in} , axial length L , layer count N_L , ferrite thickness a , and radial interlayer gap g . The current design matrix uses

$$R_{\text{in}} = 100 \text{ mm}, \quad L = 200 \text{ mm}. \quad (17)$$

The total ferrite thickness and radial occupation are

$$T_f = N_L a, \quad (18)$$

$$T_{\text{space}} = N_L a + (N_L - 1)g. \quad (19)$$

For layer i , where $i = 1$ is the innermost layer,

$$R_{i,\text{in}} = R_{\text{in}} + (i - 1)(a + g), \quad (20)$$

$$R_{i,\text{out}} = R_{i,\text{in}} + a. \quad (21)$$

The ferrite volume is calculated from the exact cylindrical-shell expression

$$V_f = \pi L \sum_{i=1}^{N_L} (R_{i,\text{out}}^2 - R_{i,\text{in}}^2). \quad (22)$$

Although V_f is displayed in mm^3 for engineering readability, all SI-derived metrics are computed after unit conversion.

For segmented-shell correction, the circumferential coverage fraction is defined as

$$\eta_\phi = 1 - \frac{N_{\text{seg}} g_\phi}{2\pi R_{\text{mid}}}, \quad (23)$$

where N_{seg} is the number of circumferential segments, g_ϕ is the slot width, and R_{mid} is the layer mid-radius. Aligned and staggered slot patterns are evaluated separately when the segmented AEDT models are available.

Fig. 2. Multilayer cylindrical shell geometry

TODO: render geometry with R_{in} , L , a , g , N_L , T_f , and T_{space} labels.

Fig. 2. Geometry definition for the multilayer thin-ferrite cylindrical shell.

G. Data Export and Paper-Use Criteria

The present main matrix is a transverse continuous-shell screening study. It is intentionally separated from axial, segmented, permeability-sensitivity, and convergence studies so that each validation stage can be tracked independently. Table II lists the current main cases.

Each shielded external-field case exports $B_{center,x}$, $B_{center,y}$, $B_{center,z}$, and $B_{center,Mag}$. The no-shield reference exports $B_{0,x}$, $B_{0,y}$, $B_{0,z}$, and $B_{0,Mag}$. Each virtual pickup-coil case exports $IntH2_{total}$ and the layerwise quantities $IntH2_{L_i}$. The post-processing scripts merge raw CSV exports by case ID and field direction and compute derived metrics only from directional components.

A case is promoted to paper-ready status only after the corresponding raw export is present, the B0 direction validation has passed, the directional field component needed for SF_q is present, and the layerwise $IntH2$ summation check has passed when layerwise $IntH2$ is used. Cases that do not satisfy these criteria remain outside the conclusion set. The vector magnitude B_{Mag} is retained only as an auxiliary consistency check and is never used as the formal shielding-factor denominator or numerator.

H. Comparison Metrics

The volume-normalized shielding index is defined as

$$\eta_S = \frac{\ln(SF_x)}{V_f}. \quad (24)$$

It is an auxiliary index introduced for this comparison, not a material property. When V_f is reported in mm^3 , the displayed unit is mm^{-3} ; SI-derived quantities are computed after converting volume to m^3 .

A dimensionless normalized form is

$$\eta_S^* = \frac{\ln(SF_x)/V_f}{\ln(SF_{x,ref})/V_{ref}}, \quad (25)$$

where the reference is the continuous single-layer shell with $a = 0.60$ mm. The loss-integral density and attenuation-normalized loss indicator are

$$\rho_H = \frac{IntH2_{total}}{V_f[m^3]}, \quad (26)$$

$$\chi_H = \frac{IntH2_{total}}{\ln(SF_x)}. \quad (27)$$

The metrics are interpreted jointly with T_{space} and the validation status. No single auxiliary metric defines a final shield geometry.

I. Traceability to Prototype SF Measurement

The FEM definitions are written so that a later laboratory shielding-factor measurement can use the same component-wise ratio. A prototype measurement would apply a calibrated field using a Helmholtz or triaxial coil, record the center-point no-shield field $B_{0,q}$ and shielded field $B_{center,q}$ using a calibrated magnetometer, and compute $SF_{q,exp} = |B_{0,q}|/|B_{center,q}|$. This traceability note is not experimental validation in the current draft.

III. VALIDATION MATRIX AND PLANNED EXPERIMENTAL CAMPAIGN

This section defines the validation campaign required before the manuscript is converted from an internal review draft to a full IEEE TIM submission. The purpose is to make each future result traceable to a specific numerical or experimental check, rather than to infer conclusions from incomplete data.

A. Mesh and Boundary-Domain Convergence

The purpose of this validation is to establish numerical robustness of SF_x and $IntH2$ for both a very thin single-layer shield and the selected four-layer candidate. The required mesh-convergence cases are C1_N1_t008_x and C2_N4_t015_g008_x. Each mesh row should record mesh level, total element count, elements across ferrite thickness, adaptive-pass count, air-domain scale, $B_{center,x}$, SF_x , $IntH2_{total}$, and relative changes with respect to the previous mesh level. The target criteria are $\Delta SF_x < 1\%$ and $\Delta IntH2_{total} < 2\%$ from the final two mesh levels. Boundary-domain convergence for C2_N4_t015_g008_x should include at least 2R, 3R, and 5R air-domain scales. This validation is designed to determine whether the current mesh and air-domain settings are sufficient for final paper use.

B. Axial Shielding Characterization

The purpose of the axial shielding study is to avoid over-generalizing transverse shielding behavior to open-ended axial shielding. The accepted B0_reference_z model provides the no-shield axial reference. The required shielded cases are C1_N1_t060_z, C2_N3_t020_g010_z, and C2_N4_t015_g008_z. The required outputs are SF_z , $LeakageRatio_z$, and SF_z/SF_x computed from the z-directed field component. The transverse ranking is not to be transferred to axial shielding unless these component-wise axial results support it.

TABLE II
MAIN CONTINUOUS-SHELL TRANSVERSE MATRIX

Case	N_L	a (mm)	g (mm)	Direction	Purpose
B0_reference_x	0	0	0	x	no-shield reference
C1_N1_t008_x	1	0.08	0	x	thinnest single layer
C1_N1_t020_x	1	0.20	0	x	thin single layer
C1_N1_t040_x	1	0.40	0	x	intermediate single layer
C1_N1_t060_x	1	0.60	0	x	equal-volume single-layer baseline
C2_N2_t020_g010_x	2	0.20	0.10	x	multilayer candidate
C2_N3_t020_g010_x	3	0.20	0.10	x	equal-volume multilayer candidate
C2_N4_t015_g008_x	4	0.15	0.08	x	equal-volume multilayer candidate

C. Segmented-Shell Correction

The purpose of the segmented-shell correction is to connect continuous-shell behavior to manufacturable segmented ferrite assemblies. The required comparison uses C2_N4_t015_g008_x as the continuous baseline and two segmented variants: aligned circumferential slots and staggered circumferential slots. The segmentation parameters are $N_{\text{seg}} = 12$ and $g_{\phi} = 1.6$ mm. The required outputs are $SF_{x,\text{seg}}/SF_{x,\text{cont}}$ and $\text{IntH2}_{\text{seg}}/\text{IntH2}_{\text{cont}}$. These quantities will determine whether continuous-shell results can support manufacturable assembly claims or should be treated only as idealized upper-bound behavior.

D. Permeability Sensitivity

The purpose of the permeability sensitivity study is to test whether the four-layer ranking depends on the screening value $\mu'_r = 1000$. The required cases are C1_N1_t060_x and C2_N4_t015_g008_x. The sweep values are $\mu'_r = 500, 1000, 2000$, and 5000 . Each row should report SF_x , $\text{IntH2}_{\text{total}}$, η_S^* , and χ_H . Ranking robustness with respect to permeability variation should be claimed only if these sweep results support it.

E. Field-Map Interpretation

The purpose of the field-map exports is to support the physical interpretation of flux redistribution and layer-resolved IntH2. The required maps are external-field B distribution, external-field flux lines, virtual-coil H_{vc} , and virtual-coil H_{vc}^2 contours. The required comparison is the equal-volume single-layer baseline C1_N1_t060_x against the four-layer candidate C2_N4_t015_g008_x. Until these maps are included in the full manuscript, interpretation of the outer-layer IntH2 contribution should remain descriptive and should not be presented as a confirmed physical mechanism.

F. Optional Prototype Measurement

The optional prototype campaign connects the FEM-defined SF_q to a laboratory measurement. The proposed setup uses a calibrated Helmholtz or triaxial coil system, a calibrated magnetometer, and center-point measurements of $B_{0,q}$ and $B_{\text{center},q}$. The desired outputs are $SF_{x,\text{exp}}$, $SF_{z,\text{exp}}$, repeatability statistics, and an uncertainty budget. Prototype data should be introduced only as experimental results if the measurement chain and uncertainty analysis are complete.

IV. EXPECTED RESULTS ORGANIZATION FOR THE FULL MANUSCRIPT

The full IEEE TIM manuscript should organize the Results section according to the validation sequence below. This organization keeps the numerical and physical claims tied to the corresponding data-entry criteria.

A. Directional Reference Validation

This subsection should report the x - and z -directed no-shield reference checks, including component purity and agreement with the expected vacuum field for the imposed excitation.

B. Transverse Continuous-Shell Characterization

This subsection should report validated x -directed continuous-shell shielding factors, leakage ratios, ferrite volume, radial occupation, and IntH2 for the transverse screening matrix.

C. Fixed-Volume Comparison

This subsection should compare the equal-volume single-layer and multilayer candidates using SF_x , $\text{IntH2}_{\text{total}}$, η_S^* , and χ_H .

D. Layer-Resolved IntH2 Distribution

This subsection should report the layerwise IntH2 fractions for multilayer cases after the layerwise summation check has passed.

E. Mesh and Boundary-Domain Convergence

This subsection should report mesh-level and air-domain-scale convergence tables and should state numerical robustness only when the acceptance criteria and metadata requirements are satisfied.

F. Axial Shielding Characterization

This subsection should report SF_z , LeakageRatio_z , and SF_z/SF_x for the selected axial cases, using B0_reference_z and z -directed shielded field components.

G. Segmented-Shell Correction

This subsection should quantify the difference between continuous and segmented shells for aligned and staggered slot patterns.

H. Permeability Sensitivity

This subsection should report the μ'_r sweep and determine whether the selected ranking remains stable across the tested permeability range.

I. Field-Map Interpretation

This subsection should use exported field maps to support the physical explanation of leakage-field attenuation and layer-resolved IntH2 distribution.

V. SUMMARY OF CURRENT DRAFT AND NEXT-STEP WORK

This internal review draft has established the TIM-oriented motivation, the component-wise directional shielding-factor definition, the no-shield reference validation logic, the virtual pickup-coil IntH2 model, the layer-resolved IntH2 decomposition, the fixed-volume comparison logic, and the validation-gated paper-use criteria. These components form the methodological core of the planned submission.

The next-step work is to complete the boundary-domain convergence study, solve and process the shielded axial cases, evaluate aligned and staggered segmented-shell models, perform the μ'_r sensitivity sweep, export field maps for physical interpretation, and optionally perform prototype shielding-factor measurements with an uncertainty budget. Results from these tasks should enter the formal manuscript only after satisfying the paper-use criteria in Section II-G.

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